

# Zachary Way

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## EDUCATION

**Georgia State University**, Atlanta, GA  
*Ph.D. in Astronomy*

*In Progress*

**University of Wisconsin – Madison**, Madison, WI  
*B.S. in Astronomy, Physics, and Mathematics*

May 2019

## EXPERIENCE

**Graduate Researcher & Teaching Assistant**  
*Georgia State University*, Atlanta, GA

Aug 2021 – Present

- Utilized low-resolution spectra from *Gaia* DR3 to constrain fundamental parameters of M dwarfs.
- 15 credit hours of teaching experience in Astronomy 1010 and 1020 labs.
- Department Webmaster since Feb 2023.
- Simulated rapidly rotating stars using the open-source 3D/2D fluid dynamic codes Rayleigh and ESTER.

**SDSS Working Group Co-Chair & Database Manager**  
*Sloan Digital Sky Survey*

May 2023 – Present

- Facilitated collaboration as co-chair of the Solar Neighborhood Census working group.
- Developed and maintain the `sdss_id` cross-match identifier spanning all SDSS iterations.

**ASAS-SN Data Aid**

May 2019 – Aug 2021

*The Ohio State University*, Columbus, OH

- Responsible for maintenance and interpretation of data produced by the All-Sky Automated Survey for Supernovae.

**MESA Research Assistant**

Nov 2017 – May 2019

*University of Wisconsin – Madison*, Madison, WI

- Worked with MESA and GYRE (stellar evolution and tidal/structure codes) under Dr. Richard Townsend.

**Undergraduate Research Assistant**

May 2017 – Sep 2017

*IceCube Neutrino Observatory*, Madison, WI

- Used Python in the calibration group analyzing LED flasher data from the South Pole neutrino detector.

## OUTREACH, AWARDS, & SERVICE

**Second Century Initiative (2CI) University Doctoral Fellow**

2021 – Present

Doctoral fellowship granted by GSU's NextGen program for core and innovative research.

**Astronomy on Tap, Atlanta – Co-Host**

2022 – Present

Organize talks, equipment, and monthly observing nights; deliver public lectures for Atlanta residents.

**Charlie Elliott Astronomy Club – Invited Lecture**

2023

Talk: *Studying Little Stars with Big Surveys*, Atlanta Astronomy Club.

**astro-ph Coffee Hour – Founder**

2021 – Present

Weekly meeting discussing recent arXiv papers at an accessible level.

**Graduate Stipends Task Force – Member**

2023

Volunteer task force for raising the minimum graduate stipend in the GSU Physics & Astronomy department.

**Polaris Mentor – The Ohio State University**

2019 – 2020

Mentored a first-year undergraduate through a semester research project for underrepresented groups.

**Breakfast of Science Champions (BoSC) – Volunteer**

2019

Taught grade-school students from underserved Columbus, OH schools about the solar system.

## PUBLICATIONS

*Refereed*

1. **Way, Z.**, Lépine, S., & Gagné, J. et al. 2026, *AJ*, 171, 252. *21,864 Unresolved, Low-mass Binaries Identified via Their Overluminosity in Gaia DR3 and a Catalog of 347,440 Systems within 100 pc of the Sun.* doi:10.3847/1538-3881/ae48ae
2. Timmermans, M., Dévora-Pajares, M., & Pozuelos, F. J. et al. (**Way, Z.**, last author) 2026, *MNRAS*, 549,

stag710. *Hidden Gems in TESS: SHERLOCK Finds Two New Rocky Planets Around Nearby M Dwarfs*. doi:10.1093/mnras/stag710

3. Qiu, D., Johnson, J. A., & Liu, C. et al. (incl. **Way, Z.**) 2026, *ApJS*, 283, 42. *Stellar Parameters of BOSS M Dwarfs in SDSS-V DR19*. doi:10.3847/1538-4365/ae40f2
4. Chandra, V., Cargile, P. A., & Ji, A. P. et al. (incl. **Way, Z.**) 2026, *ApJ*, 1000, 283. *Mapping the Distant and Metal-poor Milky Way with SDSS-V*. doi:10.3847/1538-4357/ae448a
5. Kollmeier, J. A., Rix, H.-W., & Aerts, C. et al. (incl. **Way, Z.**) 2026, *AJ*, 171, 52. *Sloan Digital Sky Survey V. Pioneering Panoptic Spectroscopy*. doi:10.3847/1538-3881/ae0576
6. Li, J., Rix, H.-W., & Ting, Y.-S. et al. (incl. **Way, Z.**) 2026, *ApJL*, 998, L33. *Variations in the Milky Way's Stellar Mass Function at  $[Fe/H] < -1$* . doi:10.3847/2041-8213/ae3d39
7. Medan, I., **Way, Z.**, & Rojas-Ayala, B. et al. 2025, *AJ*, 170, 302. *The Importance of Standardizing Spectra in the Era of Large Spectroscopic Surveys: A Case Study of M Dwarfs in SDSS-V*. doi:10.3847/1538-3881/ae0a12
8. Mészáros, S., Jofré, P., & Johnson, J. A. et al. (incl. **Way, Z.**) 2025, *AJ*, 170, 96. *SDSS-V Milky Way Mapper (MWM): ASPCAP Stellar Parameters and Abundances in SDSS-V Data Release 19*. doi:10.3847/1538-3881/ade4b9
9. Vilar, D., Souto, D., & Cunha, K. et al. (incl. **Way, Z.**) 2025, *ApJ*, 993, 241. *Chemical Abundances of M and G Dwarfs in the Hyades and Coma Berenices Open Clusters from APOGEE Spectra*. doi:10.3847/1538-4357/ae0930
10. Neumann, K. D., Holoién, T. W.-S., & Kochanek, C. S. et al. (incl. **Way, Z.**) 2023, *MNRAS*, 520, 4356. *The ASAS-SN Bright Supernova Catalogue – V. 2018–2020*. doi:10.1093/mnras/stad355
11. Christy, C. T., Jayasinghe, T., & Stanek, K. Z. et al. (incl. **Way, Z.**) 2022, *PASP*, 134, 024201. *Citizen ASAS-SN Data Release I. Variable Star Classification Using Citizen Science*. doi:10.1088/1538-3873/ac44f0
12. **Way, Z. S.**, Jayasinghe, T., & Kochanek, C. S. et al. 2022, *MNRAS*, 514, 200. *Discovery of a Highly Eccentric, Chromospherically Active Binary: ASASSN-V J192114.84+624950.8*. doi:10.1093/mnras/stac1343
13. Kawash, A., Chomiuk, L., & Strader, J. et al. (incl. **Way, Z.**) 2021, *ApJ*, 910, 120. *Classical Novae Masquerading as Dwarf Novae? Outburst Properties of Cataclysmic Variables with ASAS-SN*. doi:10.3847/1538-4357/abe53d

#### Astronomer's Telegrams

1. **Way, Z.**, Stanek, K. Z., & Kochanek, C. S. et al. 2020, *ATel*, 14122. *ASASSN-20ni: Discovery of a Likely Classical Nova in the Small Magellanic Cloud*.
2. **Way, Z.**, Desai, D., & Stanek, K. Z. et al. 2020, *ATel*, 14007. *ASAS-SN Discovery of a Luminous Star Undergoing a Deep Dimming Event*.
3. **Way, Z.**, Desai, D., & Stanek, K. Z. et al. 2020, *ATel*, 13942. *ASAS-SN Discovery of an Unusual Cataclysmic Variable (ASASSN-20jo)*.
4. **Way, Z.**, Stanek, K. Z., & Jayasinghe, T. et al. 2020, *ATel*, 13818. *ASAS-SN Discovery of a Rapid and Deep Fading Episode in a YSO Candidate*.
5. **Way, Z.**, Stanek, K. Z., & Kochanek, C. S. et al. 2019, *ATel*, 13357. *Continued Unusual Behavior of ASASSN-V J060000.76-310027.83*.
6. **Way, Z.**, Stanek, K. Z., & Kochanek, C. S. et al. 2019, *ATel*, 13346. *ASAS-SN Discovery of an Unusual, Rapidly Fading Star*.
7. **Way, Z.**, Jayasinghe, T., & Stanek, K. Z. et al. 2019, *ATel*, 13159. *ASAS-SN Discovery of a Likely RCB Variable ASASSN-V J183900.17-405428.9*.
8. **Way, Z.**, Jayasinghe, T., & Stanek, K. Z. et al. 2019, *ATel*, 13106. *ASAS-SN Discovery of an Unusual Stellar System with Deep Dimming Episodes*.

#### PRESENTATIONS

- **Talk** @ CHARA Meeting, Socorro, NM Mar 2026  
*Constraining M Giant Temperatures with Interferometric Radii*
- **Talk** @ SDSS-V Collaboration Meeting, Heidelberg, Germany Jun 2025  
*Characterizing Low-Mass Dwarfs with Alpha Hulling Standardization*
- **Talk** @ Cool Stars 22, San Diego, CA Jun 2024  
*Low-Mass, Unresolved Binaries Identified from Their Gaia XP Spectra*
- **Talk** @ American Astronomical Society #243, New Orleans, LA Jan 2024  
*Low-Mass, Unresolved Binaries Identified from Their Gaia XP Spectra*
- **Poster** @ American Astronomical Society #241, Seattle, WA Jan 2023

- Constraining Fundamental Properties of Low-Mass Dwarfs Using Low-Resolution Spectroscopy from Gaia DR3*  
 • **Talk** @ iid2022, Guntersville, AL Oct 2022  
*Constraining Fundamental Properties of Late-Type Dwarfs Using Low-Resolution Spectroscopy from Gaia DR3*
- **Talk** @ GSU Physics & Astronomy One Minute Colloquium 2021–2026  
 Annual departmental colloquium, presented each year
- **Talk** @ American Astronomical Society #237, Virtual Jan 2021  
*Discovery of a Highly Eccentric, Chromospherically Active Binary System: ASASSN-V J192114.84+624950.8*
- **Poster** @ UW-Madison Undergraduate Research Symposium, Madison, WI Apr 2019  
*Interpolation Schemes that Don't Ruin Stars*

#### TECHNICAL SKILLS

**Languages:** Python, Fortran, Mathematica, Bash, PostgreSQL

**Software:** L<sup>A</sup>T<sub>E</sub>X, Linux, SAOImage DS9, Jupyter, MATPLOTLIB, MESA, GYRE, PHOEBE, ASTROPY, EMCEE

**General:** Version Control, Research Presentation, Numerical Methods, Building Mathematical Models